Default of Credit Card Clients

BAN 620 – Data Mining Group Project

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Abstract

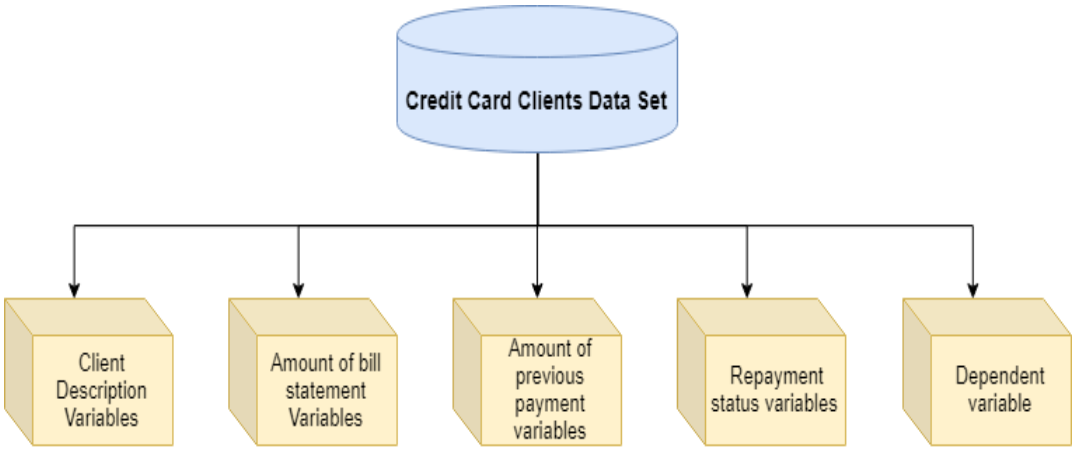
Credit Risk Assessment has become pivotal among financial institutions to avoid financial risk and reduce the count of defaulters globally. A credit risk is the risk of default on a credit card payment that may arise from a borrower/credit card user failing to make required payments. In this process the credit history of the customer undergoes a rigorous check to ensure that whether they are capable enough to pay off the credit amount. The objective is to make a prediction whether the customer will pay his debt timely or not and then decide on a cut-off criterion.

Goal

- Predict the Credit Card Payment Defaulters and Non-Defaulters for the next month.
- Aim is to find/build a Data Mining model that will help us determine defaulters with high accuracy.
- Find potential customers for the institute/banks who can make the credit card payments on time.
- Classify potential customers who will default according to Profiling and Propensity.

Data Set Summary

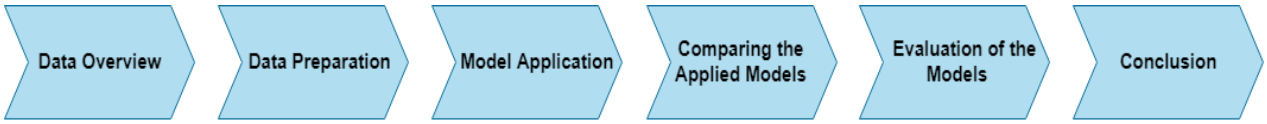
The data set is of a Taiwanese Financial firm dated from April 2005 to September 2005, including 30000 unique customers (rows). It has 25 variables including the dependent variable default.payment.next.month (June 2005).The data set can be portrayed as combination of 5 parts .



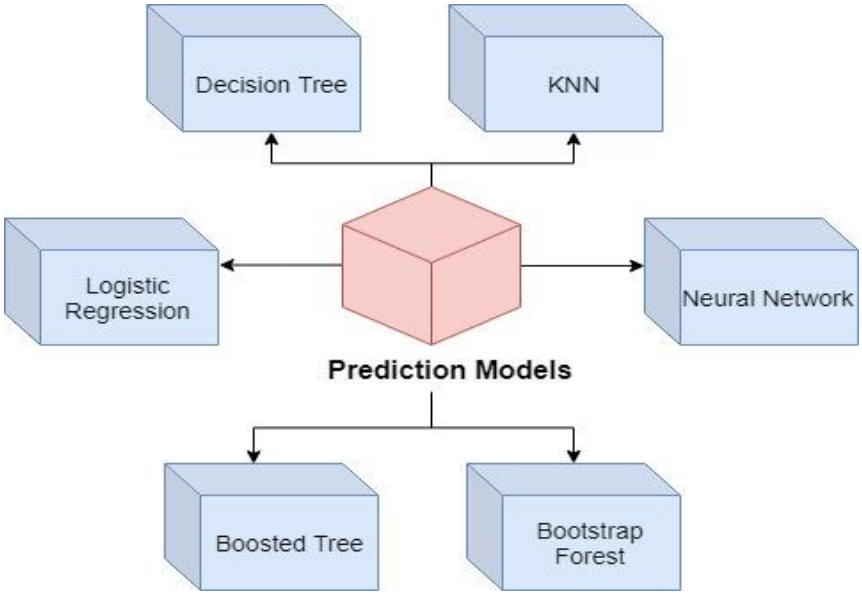
Variable	Description	Variable Segment
ID	ID of each client	Client Description Variables
LIMIT_BAL	Amount of given credit in NT dollars (includes individual and family/supplementary credit)	
SEX	Gender (1=male, 2=female)	
EDUCATION	(1=graduate school, 2=university, 3=high school, 4=others)	
MARRIAGE	Marital status (1=married, 2=single, 3=others)	
AGE	Age in years	
PAY_0	Repayment status in September, 2005 (-1=pay duly, 1=payment delay for one month2=for 2 months, ...)	Repayment status variables
PAY_2	Repayment status in August, 2005 (scale same as above)	
PAY_3	Repayment status in July, 2005 (scale same as above)	
PAY_4	Repayment status in June, 2005 (scale same as above)	
PAY_5	Repayment status in May, 2005 (scale same as above)	
PAY_6	Repayment status in April, 2005 (scale same as above)	
BILL_AMT1	Amount of bill statement in September, 2005 (NT dollar)	Amount of bill statement Variables
BILL_AMT2	Amount of bill statement in August, 2005 (NT dollar)	
BILL_AMT3	Amount of bill statement in July, 2005 (NT dollar)	
BILL_AMT4	Amount of bill statement in June, 2005 (NT dollar)	
BILL_AMT5	Amount of bill statement in May, 2005 (NT dollar)	
BILL_AMT6	Amount of bill statement in April, 2005 (NT dollar)	
PAY_AMT1	Amount of previous payment in September, 2005 (NT dollar)	Amount of previous payment variables
PAY_AMT2	Amount of previous payment in August, 2005 (NT dollar)	
PAY_AMT3	Amount of previous payment in July, 2005 (NT dollar)	
PAY_AMT4	Amount of previous payment in June, 2005 (NT dollar)	
PAY_AMT5	Amount of previous payment in May, 2005 (NT dollar)	
PAY_AMT6	Amount of previous payment in April, 2005 (NT dollar)	
default.payment.next.month	Default payment in June, 2005 (1=yes, 0=no)	Dependent Variable

Process Design

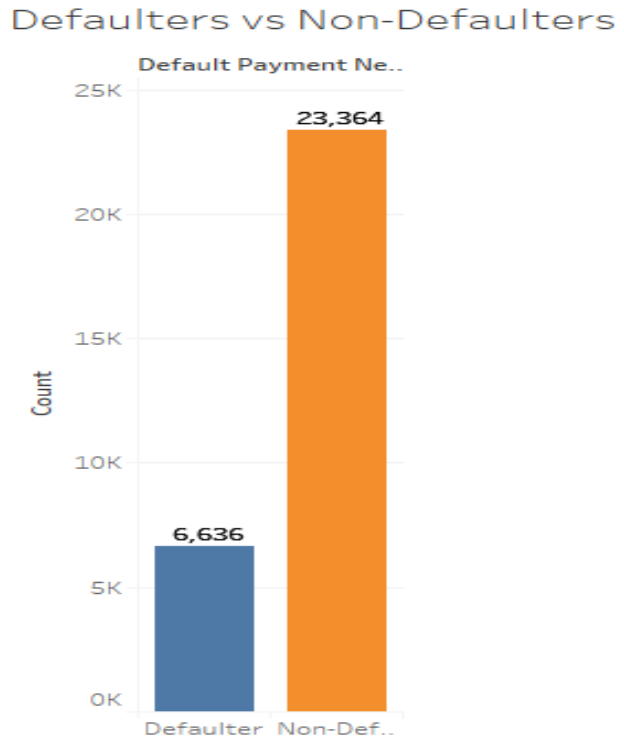
- The process is done by Supervised learning Algorithm
- The idea behind using this is we have a prior knowledge on our output values
- It acts as a guide to teach the algorithms what conclusion it should come up with
- Proposed algorithms are Decision Tree, Logistic Regression, Boosted Tree, Bootstrap Forest, Neural Network, K-NN.



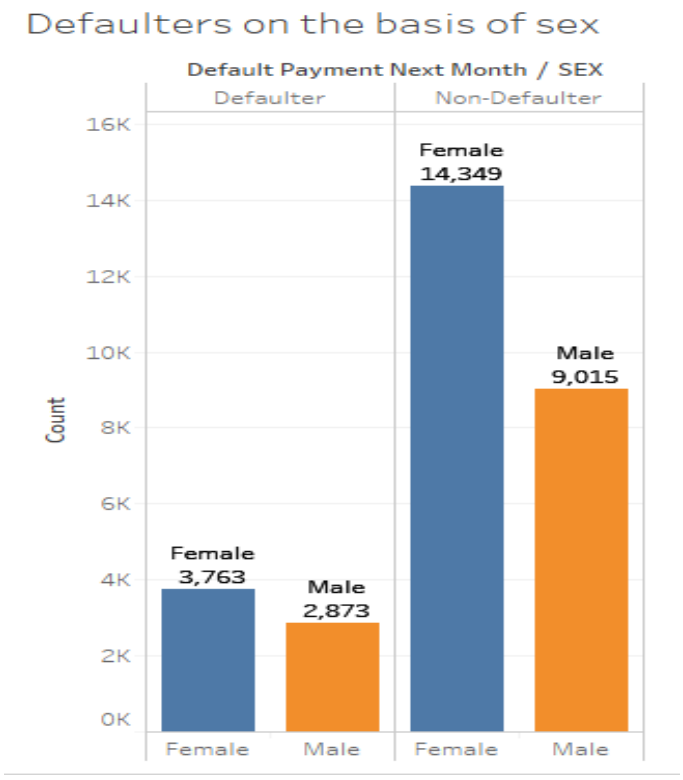
Process Approach



Exploratory Data Analysis:



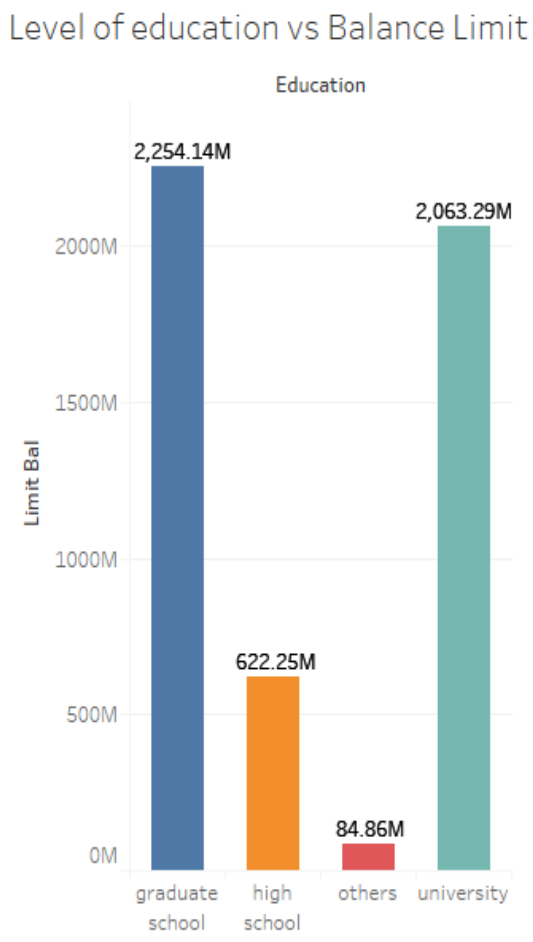
(a)



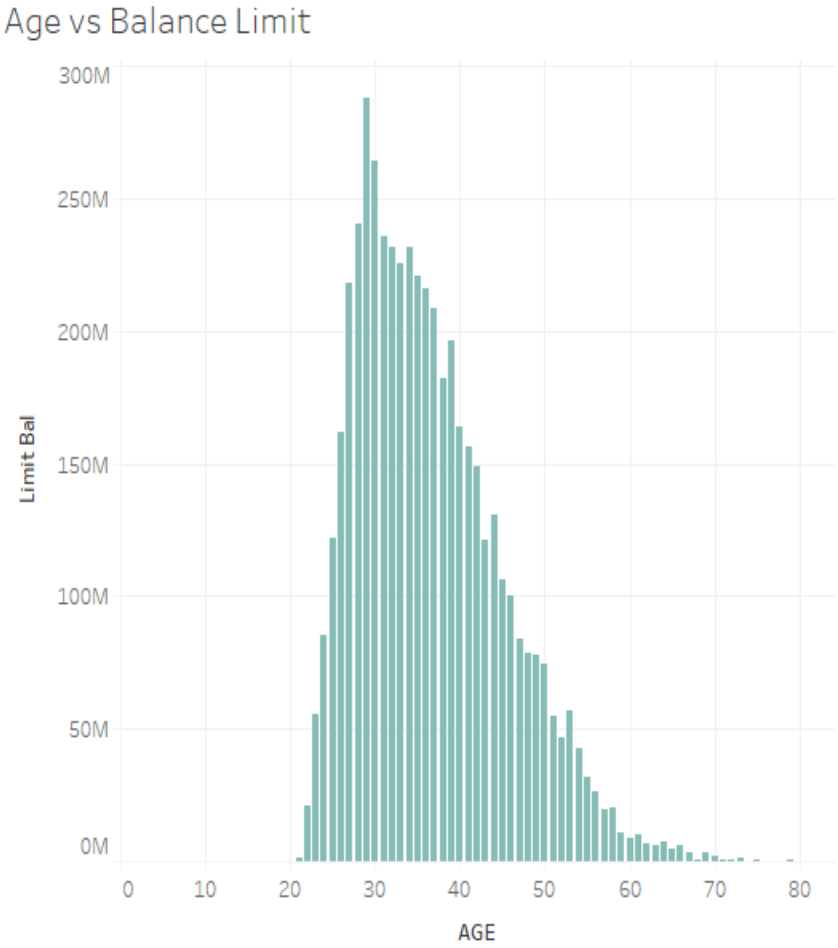
(b)

Report a -> Out of 30000 unique customers 6636 are defaulters wile 23364 are non-defaulters which gives a percentile ratio of 22% defaulters: 88% non-defaulters hence we can say that that data is not biased to one category.

Report b->It gives the overview about the split on male and female. As we can see there are 60% of the data consist of women while 56% of defaulters are women.



(c)



(d)

Report c -> We can see our dataset follows global data trends I.e. higher the level of education higher is the balance limit.

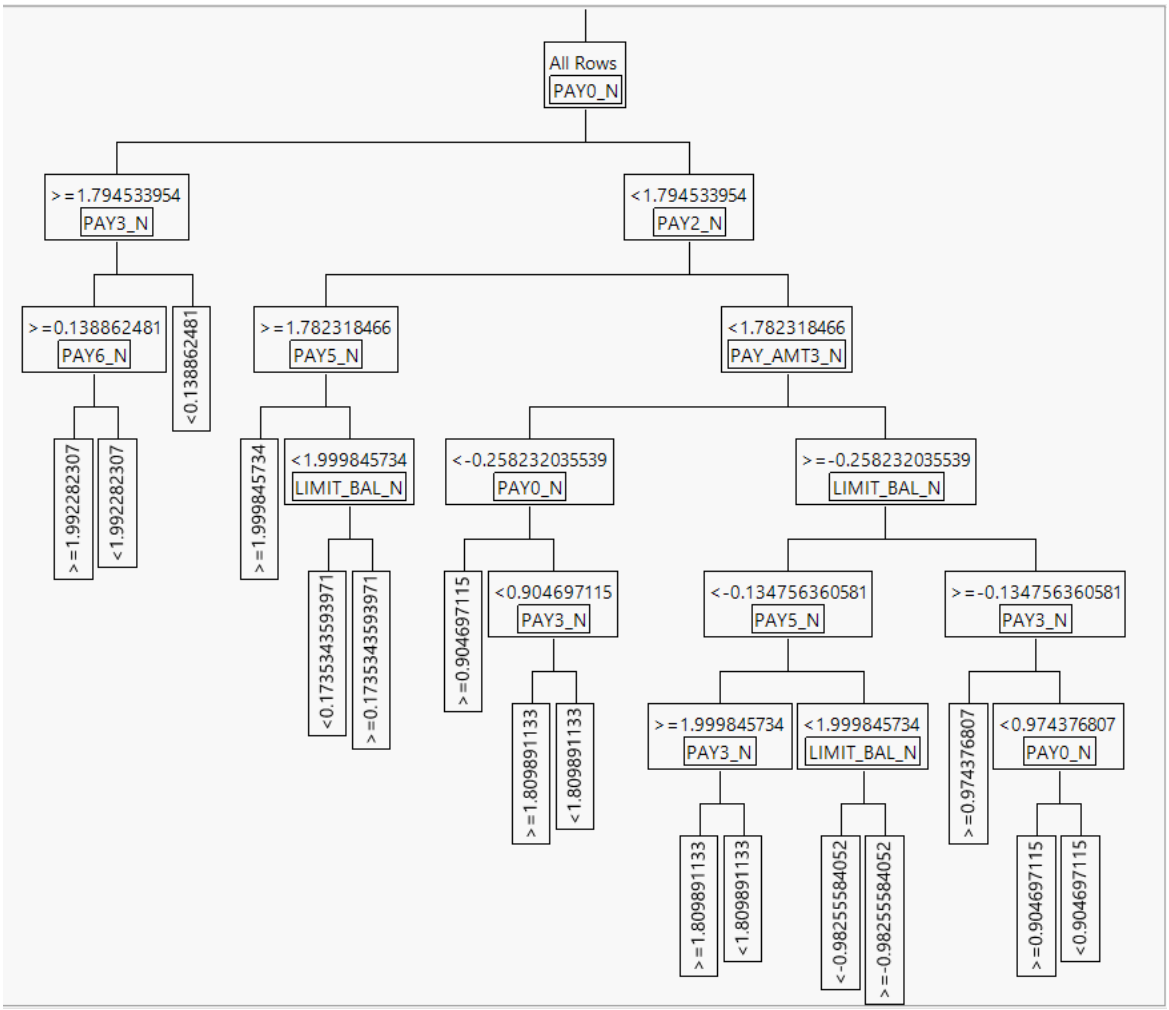
Report d -> we can see the graph is of a form of bell curve, as the age increases it is natural the debt repayment capability of an individual increases as his salary/designation increases. As the individual nears his retirement gradually his debt repayment will decrease because of the time factor.

Main Chapter

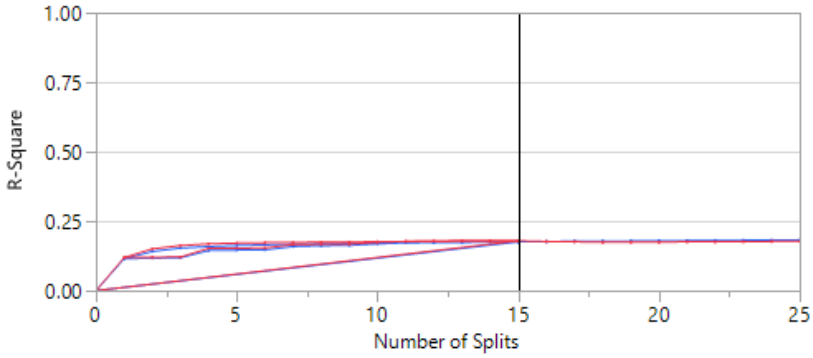
Model 1: Decision Tree:

We created a Decision Tree model, we took default payment (Yes = 1, No = 0), as the response variable. We obtained the following results using the Fit Details, Leaf Report, Column Contribution, ROC curve, Lift curve and Profiler.

- This is the small tree view which show all the splits of the Model.



Split History



Validation Data in Red

Fit Details

Measure	Training	Validation	Definition
Entropy RSquare	0.1759	0.1794	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.2603	0.2643	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.4367	0.4318	∑ -Log(ρ[j])/n
RMSE	0.3696	0.3670	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2732	0.2708	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.1799	0.1774	∑ (ρ[j]≠ρMax)/n
N	17875	12125	n

Confusion Matrix

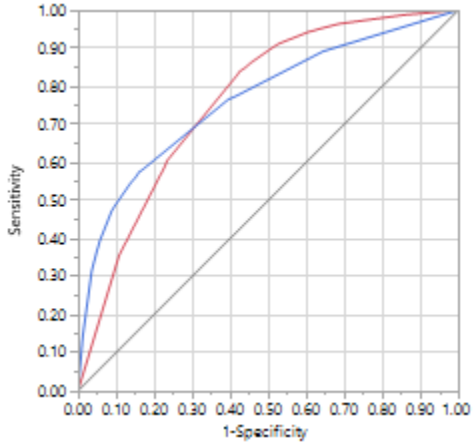
Training

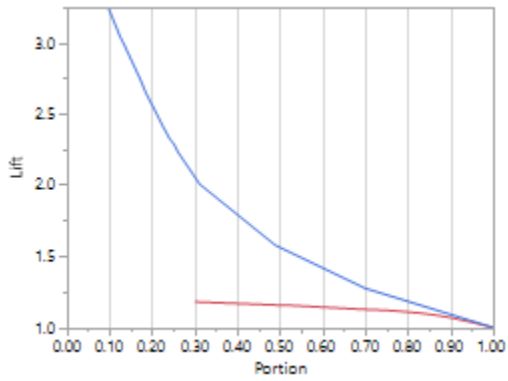
Actual	Predicted Count	
Y	no	yes
no	13191	709
yes	2507	1468

Validation

Actual	Predicted Count	
Y	no	yes
no	8995	469
yes	1682	979

- The Fit details of the Decision tree model are seen above. As seen in the results, we observe a RSquare value for training and validation data at 0.2603 and 0.2643 respectively, which tell us that this model does not suffer from overfitting. Moreover, we see that there were a total 15 splits used for the final Decision tree. Moreover, we observe that Training and Validation data RMSE values are at 0.3696 & 0.3670; which will be used for model performance comparisons. Misclassification rate is 17.74% which is less.
- From the split history, we can see that the validation data (red line) is split with n = 15 as this represents the most optimal split value which was chosen due to its highest R^2 value. It is seen that although the number of splits remains same for the training data.
- Confusion Matrix of Validation states
    - 8995 entries are predicated non defaulters which are actually non defaulter
    - 1682 entries are predicted non defaulters which are actually defaulters
    - 469 entries are predicted defaulters which are actually non defaulters
    - 979 entries are predicted defaulter which are actually defaulters
    - [(8995+978)/ 8995+978+469+1682] = 82.26% entries are predicted correctly



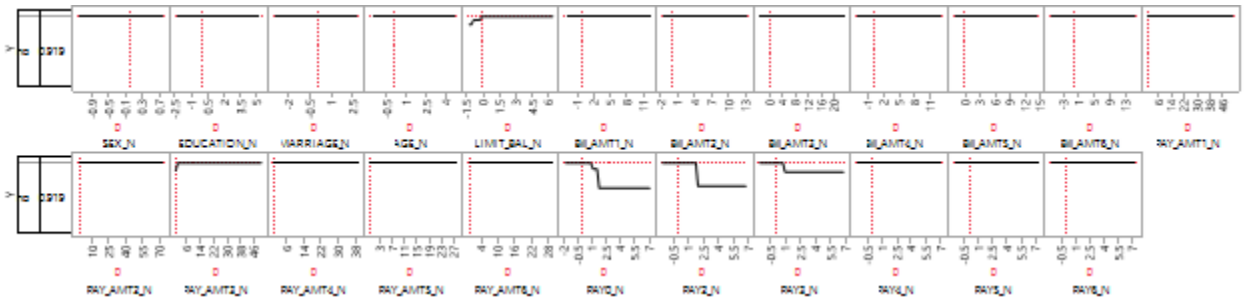


ROC curve:

- ROC curve tells us whether model is statistically significant or vice versa.
- As we can see from the Curve below, Curve is above diagonal.
- We can say that model is accurate and statistically significant

Lift curve

- From the lift curve we can say that our model predicts the defaulters 3 times better than base line or output which we get after running the random sample data.



- The Prediction Profiler tells us the relationship of input variables with response variable.
- Eg- SEX_N, EDUCATION_N, AGE_N has linear relationship with response variable Y.
- PAY0_N has no relationship with response variable Y

Model 2: Boosted Tree:

Boosted tree is improved model of Regression Tree. Like a bootstrap forest, a boosted tree is a small tree that is repeatedly fit to the data using a random sample of predictors. The difference is that boosted trees make use of residuals based on misclassifications or errors which are reduced until an optimal tree is generated. From the snippet below we can see that a small tree was used with 3 splits and 50 layers were used for the model.

Overall Statistics

Measure	Training	Validation	Definition
Entropy RSquare	0.1919	0.1835	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.2816	0.2698	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.4282	0.4297	∑ -Log(ρ[j])/n
RMSE	0.3655	0.3659	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2724	0.2723	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.1761	0.1758	∑ (ρ[j]≠ρMax)/n
N	17875	12125	n

Confusion Matrix

Training

Actual	Predicted Count	
Y	no	yes
no	13272	628
yes	2519	1456

Validation

Actual	Predicted Count	
Y	no	yes
no	9041	423
yes	1708	953

- The R square of Boosted tree is 26.98% which is slightly better than Decision tree.
- RMSE is 0.3659 which is less than Decision tree as well.
- Misclassification tree is 17.58% which is also less than Decision tree

Model 3: K – Nearest Neighbor

k-Nearest Neighbors method (k-NN) can be used for Classification of a categorical outcome or Prediction of a numerical outcome. To classify or predict a new record, this method relies on finding "similar" records in the training data. k-NN is a nonparametric method. Does not make assumptions about the form of the relationship between the class membership (Y) and the predictors X1, X2, … Xp. Does not involve estimation of parameters in an assumed function form (e.g., linear relationship in regression) Characteristics are:

- Data driven, not model driven (like multiple regression)
- No assumptions about the data relationship
- Conceptually simple/straightforward
- Highly automated

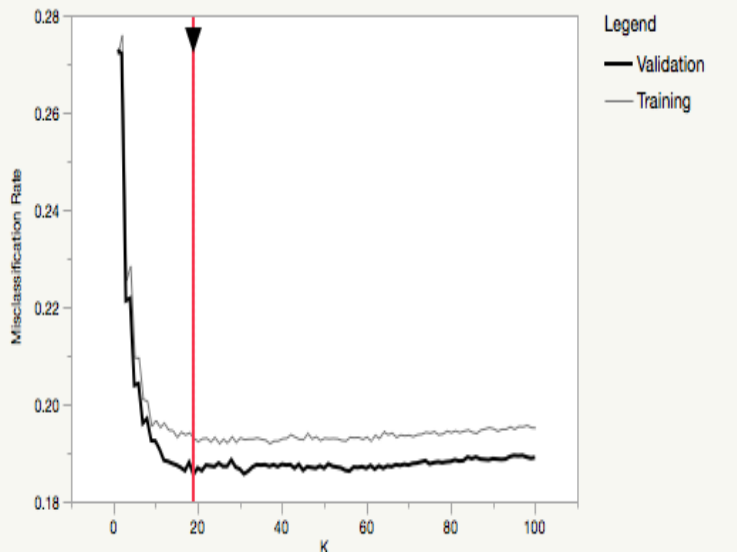
Model Selection

- Output below is of model selection.
- Model selection gives us the least classification rate with selection of optimal k
- Here Red line represents the Validation where 19 is the optimal k with least classification rate

K Nearest Neighbors

Y

Model Selection



Training

K	Count	Misclassification Rate	Misclassifications
1	17875	0.27189	4860
2	17875	0.27608	4935
3	17875	0.22557	4032
4	17875	0.22859	4086
5	17875	0.20973	3749
6	17875	0.20962	3747
7	17875	0.20117	3596
8	17875	0.20073	3588
9	17875	0.19558	3496
10	17875	0.19687	3519

Validation

K	Count	Misclassification Rate	Misclassifications
13	12125	0.18821	2282
14	12125	0.18779	2277
15	12125	0.18746	2273
16	12125	0.18697	2267
17	12125	0.18631	2259
18	12125	0.18804	2280
19	12125	0.18548	2249
20	12125	0.18689	2266
21	12125	0.18631	2259
22	12125	0.18746	2273

Confusion Matrix for Best K=19

Training

Actual Y	Predicted Count no	Predicted Count yes
no	13197	703
yes	2749	1226

Validation

Actual Y	Predicted Count no	Predicted Count yes
no	9021	443
yes	1806	855

- The misclassification rate is 18.54% at optimal k = 19.
- The misclassification rate is little more compared to decision tree and boosted tree

Model 4: Fit Nominal Logistic:

We created a Fit nominal Logistic model, we took default payment (Yes = 1, No = 0), as the response variable. We obtained the following results using the Fit Details, Effect Summary table, Parameter Estimate, Confusion Matrix

- Effect Summary Table shows us the parameter Contribution to the model
- As we can see PAY0_N has contributed the most in the prediction model
- The parameters who contributed are Pay_amt1_n, bill_amt1_1
- Bill_amt6, Bill_amt5, py6_n has least contribution to the model.

Fit Details

Measure	Training	Validation	Definition
Entropy RSquare	0.1201	0.1197	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.1829	0.1818	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.4663	0.4633	∑ -Log(ρ[j])/n
RMSE	0.3814	0.3794	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2954	0.2938	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.1905	0.1877	∑ (ρ[j]≠ρMax)/n
N	17875	12125	n

Confusion Matrix

Training

Actual	Predicted Count	
Y	no	yes
no	13537	363
yes	3043	932

Validation

Actual	Predicted Count	
Y	no	yes
no	9213	251
yes	2025	636

- The Rquare of this model is 18.18% which is less compared to other models
- RMSE is 0.3794 which is more compared to other models
- The misclassification rate is 18.77% which is more compared to other models

Model 5: Neural Networks

We implemented the model with neural network which performs well for both classification and prediction. A neural network model enables capture of complex relationships between predictors and response variables. After running various tuning parameters for the neural network, we have found that the best performance is attained when using 20 nodes for the hidden layer for each activation function TanH, Linear and Gaussian. We observe that we have lower RMSE values for both training and validation with increased RSquare values for both training and validation data. This lower RMSE for validation data tells us that the neural network performs the best as compared to other models.

Model NTanH(3)

Training

Y

Measures	Value
Generalized RSquare	0.2704469
Entropy RSquare	0.1834981
RMSE	0.3680352
Mean Abs Dev	0.2706409
Misclassification Rate	0.1797483
-LogLikelihood	7733.8876
Sum Freq	17875

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	13180	720
yes	2493	1482

Validation

Y

Measures	Value
Generalized RSquare	0.270447
Entropy RSquare	0.1839817
RMSE	0.3663009
Mean Abs Dev	0.2690742
Misclassification Rate	0.1772371
-LogLikelihood	5206.6226
Sum Freq	12125

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	8991	473
yes	1676	985

Model NTanH(10)

Training

Y

Measures	Value
Generalized RSquare	0.2819638
Entropy RSquare	0.1921635
RMSE	0.3663936
Mean Abs Dev	0.2679412
Misclassification Rate	0.1794685
-LogLikelihood	7651.8088
Sum Freq	17875

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	13243	657
yes	2551	1424

Validation

Y

Measures	Value
Generalized RSquare	0.2707323
Entropy RSquare	0.1841959
RMSE	0.366172
Mean Abs Dev	0.2669605
Misclassification Rate	0.1769897
-LogLikelihood	5205.2563
Sum Freq	12125

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	9037	427
yes	1719	942

Model NTanH(20)NLinear(20)NGaussian(20)NBoost(5)

Training

Y

Measures	Value
Generalized RSquare	0.3053075
Entropy RSquare	0.2099756
RMSE	0.3625136
Mean Abs Dev	0.2670411
Misclassification Rate	0.1772308
-LogLikelihood	7483.0932
Sum Freq	17875

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	13250	650
yes	2518	1457

Validation

Y

Measures	Value
Generalized RSquare	0.2763605
Entropy RSquare	0.1884309
RMSE	0.3654157
Mean Abs Dev	0.2682287
Misclassification Rate	0.1766598
-LogLikelihood	5178.2348
Sum Freq	12125

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	9025	439
yes	1703	958

Model NTanH(10)NLinear(10)NGaussian(10)NTanH2(10)NLinear2(10)NGaussian2(10)

Training

Y

Measures	Value
Generalized RSquare	0.2937294
Entropy RSquare	0.2010991
RMSE	0.364342
Mean Abs Dev	0.265654
Misclassification Rate	0.1760559
-LogLikelihood	7567.1713
Sum Freq	17875

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	13269	631
yes	2516	1459

Validation

Y

Measures	Value
Generalized RSquare	0.2721934
Entropy RSquare	0.1852935
RMSE	0.3660231
Mean Abs Dev	0.266319
Misclassification Rate	0.1766598
-LogLikelihood	5198.2528
Sum Freq	12125

Confusion Matrix

Actual	Predicted Count	
Y	no	yes
no	9018	446
yes	1696	965

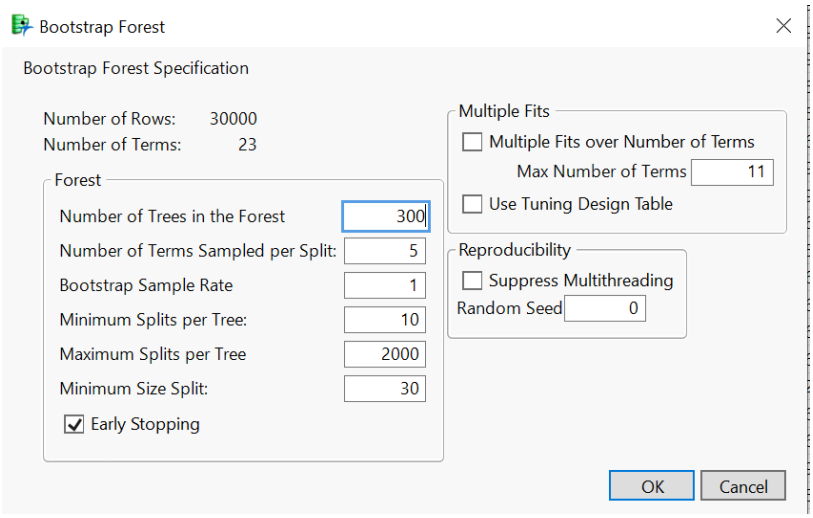
- The R square of Neural Network model with 20 nodes for the hidden layer for each activation function TanH, Linear and Gaussian is 27.63% which is slightly better than all the models mentioned above.
- RMSE is 0.3654 which is less than other models.
- Misclassification tree is 17.66% which is also less.

Model 6: Bootstrap Forest

The Bootstrap Forest platform fits an ensemble model by averaging many decision trees each of which is fit to a bootstrap sample of the validation data. Each split in each tree considers a random subset of the predictors.

The specifications of bootstrap forest is mentioned below

We selected no. of trees in forest = 300, no. of terms samples per split = 5 and sample rate = 1



Overall Statistics

Measure	Training	Validation	Definition
Entropy RSquare	0.3329	0.1971	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.4550	0.2878	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.3535	0.4225	∑ -Log(ρ[j])/n
RMSE	0.3329	0.3635	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.2446	0.2661	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.1695	0.1757	∑ (ρ[j]≠ρMax)/n
N	17875	12125	n

Confusion Matrix

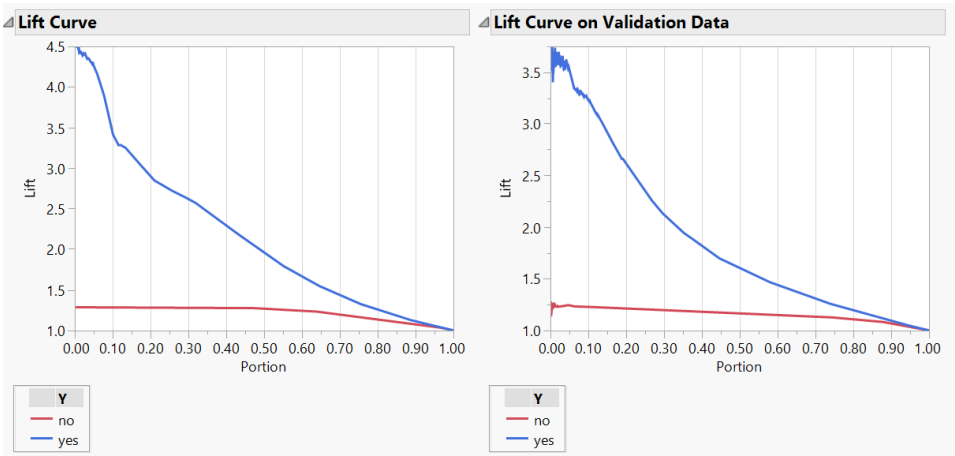
Training

Actual	Predicted Count	
Y	no	yes
no	13344	556
yes	2474	1501

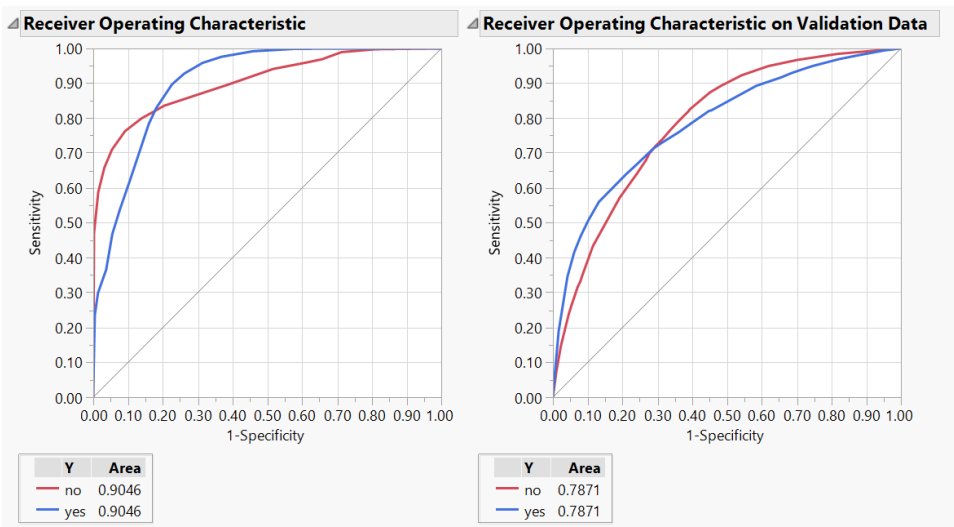
Validation

Actual	Predicted Count	
Y	no	yes
no	9052	412
yes	1718	943

- The R square of Bootstrap forest model is 28.78% which is better than all the models mentioned above.
- RMSE is 0.3635 which is least compared to other models.
- Misclassification tree is 17.57% which is also least compared to other models.



From the lift curve we can say that our model predicts the defaulters 3.5 times better than base line or output which we get after running the random sample data.



- ROC curve tells us whether model is statistically significant or vice versa.
- As we can see from the Curve below, Curve is above diagonal.
- We can say that model is accurate and statistically significant

Model Comparison

Measures of Fit for Y

Creator	.2 .4 .6 .8	Entropy RSquare	Generalized RSquare	Mean -Log p	RMSE	Mean Abs Dev	Misclassification Rate	N
Partition		0.1794	0.2643	0.4318	0.3670	0.2708	0.1774	12125
Boosted Tree		0.1835	0.2698	0.4297	0.3659	0.2723	0.1758	12125
Bootstrap Forest		0.1965	0.2870	0.4228	0.3636	0.2663	0.1757	12125
K Nearest Neighbors		.	.	.	.	.	0.1855	12125
Fit Nominal Logistic		0.1197	0.1818	0.4633	0.3794	0.2938	0.1877	12125
Neural		0.1840	0.2704	0.4294	0.3663	0.2691	0.1772	12125
Neural		0.1842	0.2707	0.4293	0.3662	0.2670	0.1770	12125
Neural		0.1884	0.2764	0.4271	0.3654	0.2682	0.1767	12125
Neural		0.1853	0.2722	0.4287	0.3660	0.2663	0.1767	12125

Since the required prediction is a classification problem the three main criteria for model selection are

- Low Misclassification Rate
- Low RMSE
- High Rsquare
- No overfitting

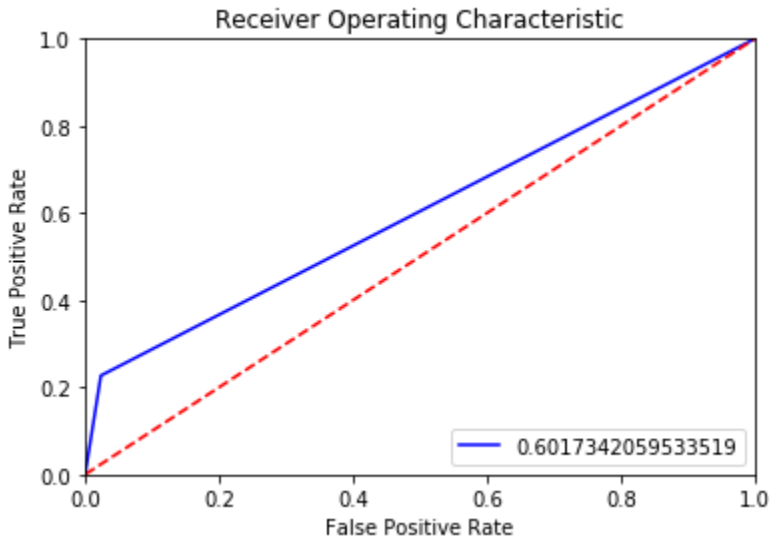
Bootstrap forest follows all the four criteria's and hence it is the best model.

Tuning the Model

- Type 2 (Applicant will not pay the loan) Accuracy Rate for Bootstrap forest = 35%
- As You can see the overall misclassification rate is 93% but the downside is type 2 accuracy is also low
- Hence to improve it we have tuned the model based on p value in python using spyder, following results were obtained

```
from sklearn.metrics import confusion_matrix,accuracy_score
cfm=confusion_matrix(Y_test.tolist(),y_pred_class)#we are tycasting to lis
print(cfm)
accuracy_score=accuracy_score(Y_test.tolist(),y_pred_class)
print("Accuracy of the model:",accuracy_score)
#%%
for a in np.arange(0,1,0.05):
    predict_mine = np.where(y_pred_prob[:,0]<a,1,0)
    cfm=confusion_matrix(Y_test.tolist(),predict_mine)
    total_err=cfm[0,1]+cfm[1,0]
    print("Errors at treshold",a,":",total_err,"type 2 error :",cfm[1,0])
#%% auc
    from sklearn import metrics
fpr, tpr, threshold = metrics.roc_curve(Y_test, Y_pred)
auc = metrics.auc(fpr,tpr)
print(auc)
    #%% roc
import matplotlib.pyplot as plt
plt.title('Receiver Operating Characteristic')
plt.plot(fpr, tpr, 'b', label = auc)
plt.legend(loc = 'lower right')
plt.plot([0, 1], [0, 1],'r--')
plt.xlim([0, 1])
plt.ylim([0, 1])
plt.ylabel('True Positive Rate')
plt.xlabel('False Positive Rate')
plt.show()
```

```
Errors at treshold 0.0 : 2018 type 2 error : 2018
Errors at treshold 0.05 : 2014 type 2 error : 2011
Errors at treshold 0.1 : 2011 type 2 error : 2005
Errors at treshold 0.15000000000000002 : 2006 type 2 error : 1997
Errors at treshold 0.2 : 1986 type 2 error : 1973
Errors at treshold 0.25 : 1965 type 2 error : 1949
Errors at treshold 0.30000000000000004 : 1944 type 2 error : 1921
Errors at treshold 0.35000000000000003 : 1918 type 2 error : 1884
Errors at treshold 0.4 : 1866 type 2 error : 1804
Errors at treshold 0.45 : 1790 type 2 error : 1685
Errors at treshold 0.5 : 1724 type 2 error : 1560
Errors at treshold 0.55 : 1646 type 2 error : 1388
Errors at treshold 0.6000000000000001 : 1661 type 2 error : 1267
Errors at treshold 0.65 : 1701 type 2 error : 1171
Errors at treshold 0.7000000000000001 : 1786 type 2 error : 1086
Errors at treshold 0.75 : 2209 type 2 error : 883
Errors at treshold 0.8 : 3453 type 2 error : 600
Errors at treshold 0.8500000000000001 : 4706 type 2 error : 397
Errors at treshold 0.9 : 5748 type 2 error : 171
Errors at treshold 0.9500000000000001 : 6598 type 2 error : 31
```




```
[[5138 1844]
 [ 779 1239]]
Accuracy of the model: 0.7085555555555556
```

Results: Though the accuracy was reduced to 71%, the model got trained well on type 2 errors improvising the type 2 accuracy rate to 61% and reducing the type 1 accuracy to smaller extent of 74%.

Conclusion

To summarize we can conclude that bootstrap forest marginally outperformed other models in terms of misclassification rate, RMSE, R square and overfitting criterion. Also, the model was tuned so that it can classify type 2 errors more efficiently with an overall accuracy of 71%.

We can conclude that thus Taiwanese financial institutions can use the Bootstrap forest model to reduce credit risk in there by safeguarding their organizations from potential risks.

Bibliography

- https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients#
- Data Mining for Business Analytics – Concepts, Techniques, and Applications with JMP PRO, by Galit Shmueli, Peter C. Bruce, Mia L. Stephens, Nitin R. Patel

Appendix

Variable Name	Description
Activation function	Abstraction representing the rate of action potential firing in the cell
Biased	Belonging to particular type(defaulter/non-defaulter)
Confusion Matrix	Matrix that helps to find accuracy
Credit Risk	Chance of Default
Categorical	Data can be measured (usually 0,1,2,3.... or Yes/No)
Classification	Classify into classes
Cut-off	Criteria to give the individual card credit or not
Data driven	Determined by or dependent on the collection or analysis of data.
Defaulter	The individual to fails to repay his credit card bill
ensemble	Group of items viewed as a whole rather than individually.
Exploratory	Mining Data trends
Financial Risk	Potential Risk due to defaulter
Global data trends	Data trends that are accepted globally
Model	Prediction Model
Neural	Relating to a nerve or the nervous system.
Predictors	Independent variables
Profiling	The recording and analysis of a person's psychological and behavioral characteristics
Propensity	An inclination or natural tendency to behave in a particular way.
ROC	Receiver Operating Characteristics
Regression	A measure of the relation between the mean value of one variable and corresponding values of other variables
Supervised	We know the output criteria (Yes/No)
Statistically significant	Supports our cause
Snippet	Screenshot
Tuning	Improve the accuracy
Training	Data that our model trains on
Validation	Data on which our model checks its accuracy